

CHICAGO MIDWAY, IL, USA

WMO: 725340

Lat: 41.786N

Lon: 87.752W

Elev: 612

StdP: 14.37

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 14819

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.0	5.0	-10.0	3.3	2.3	-5.8	4.1	6.5	26.0	29.9	24.2	28.8	11.3	270	0.562

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.7	91.8	74.6	89.5	73.2	86.8	72.0	77.8	87.9	76.0	85.2	74.4	83.3	12.4	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.8	133.2	84.0	72.9	124.9	81.9	71.7	119.7	80.7	41.7	88.1	39.8	85.5	38.3	83.2	85.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.3	20.9	19.0	-6.7	96.6	6.8	3.9	-11.7	99.4	-15.7	101.7	-19.5	103.8	-24.5	106.6
			-7.6	80.5	6.5	2.4	-12.3	82.2	-16.1	83.7	-19.8	85.0	-24.6	86.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.1	25.5	29.2	39.3	50.5	61.2	71.3	76.1	74.5	67.7	55.1	42.0	31.1
	DBStd	19.84	11.67	11.19	10.84	9.32	9.13	7.43	5.75	5.21	7.64	8.94	9.70	10.75
	HDD50	2738	760	586	366	103	10	0	0	0	48	273	592	
	HDD65	5814	1224	1002	799	444	188	26	1	3	58	329	689	1051
	CDD50	3497	1	4	34	118	357	640	808	758	532	205	35	5
	CDD65	1100	0	0	3	10	70	216	344	296	140	21	0	0
	CDH74	9339	0	0	20	103	632	1937	3183	2268	1049	147	0	0
	CDH80	3150	0	0	2	19	192	688	1190	703	327	29	0	0
	WSAvg	9.9	11.0	11.0	11.0	11.3	10.0	8.9	8.6	8.0	8.6	9.7	10.4	10.3
	PrecAvg	35.7	2.0	1.9	2.3	3.3	4.1	4.1	3.8	4.0	2.9	3.0	2.5	1.9
(15) Precipitation	PrecMax	46.3	4.6	6.1	5.0	7.3	7.5	10.5	8.1	9.0	10.3	8.4	6.0	4.6
	PrecMin	22.5	0.5	0.5	0.8	1.3	0.4	1.0	1.7	1.1	0.5	0.9	0.3	0.5
	PrecStd	5.4	1.1	1.1	1.1	1.4	1.7	2.0	1.6	1.9	2.0	1.9	1.5	0.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.1	63.2	77.0	82.3	89.8	93.3	97.6	93.4	91.8	84.1	70.3	62.5
		MCWB	54.8	53.5	63.1	63.7	70.3	75.0	78.4	75.6	73.2	68.4	58.0	59.1
	2%	DB	50.5	55.4	68.6	76.6	85.1	90.4	92.6	90.0	87.8	78.5	64.4	55.2
		MCWB	45.8	48.6	57.3	60.9	68.2	73.0	76.4	73.8	71.2	64.0	56.5	50.7
	5%	DB	44.8	48.8	61.7	71.9	81.3	87.6	89.5	86.6	83.6	73.3	61.2	50.1
		MCWB	41.0	43.5	53.2	58.0	66.0	72.3	74.3	72.6	69.0	61.9	54.3	46.1
	10%	DB	40.4	43.6	55.4	66.1	76.7	83.9	86.4	83.8	80.5	69.6	56.6	45.3
		MCWB	37.4	39.5	47.8	54.7	64.1	70.3	72.5	71.4	67.3	60.1	50.1	41.4
	0.4%	WB	55.3	56.5	63.8	67.9	73.7	78.8	81.9	79.1	76.0	71.0	62.2	59.3
		MCDB	57.9	60.6	75.6	78.8	85.4	89.4	92.2	90.3	87.1	81.4	66.0	61.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	47.0	49.8	59.2	63.1	70.8	76.2	78.5	76.8	73.5	67.5	58.8	52.2
		MCDB	49.2	55.8	66.2	72.3	81.0	85.9	88.6	85.5	83.2	74.3	63.7	54.7
	5%	WB	41.2	44.1	54.2	59.6	68.3	74.0	76.6	75.0	71.3	64.2	54.6	46.2
		MCDB	44.2	48.3	61.0	69.9	77.6	83.4	86.4	82.8	79.9	70.8	60.1	49.0
	10%	WB	37.3	39.5	48.4	55.9	65.8	72.1	74.8	73.5	69.3	61.1	50.8	41.4
		MCDB	40.2	43.5	55.5	65.1	74.5	81.5	83.9	81.1	77.7	68.5	55.8	45.0
(35) Mean Daily Temperature Range	5% DB	MDBR	12.5	13.2	14.8	17.1	17.6	16.9	15.7	14.7	16.1	15.4	13.3	11.5
		MCDBR	16.6	19.5	22.7	25.3	22.4	19.7	18.2	18.0	19.8	21.3	18.4	15.9
		MCWBR	14.7	15.4	15.2	14.6	11.0	9.2	7.8	7.8	8.6	12.0	13.3	13.7
	5% WB	MCDBR	16.1	18.1	21.2	23.0	20.2	18.2	17.1	15.5	16.7	17.9	17.0	14.7
		MCWBR	14.8	15.6	16.1	15.8	12.1	10.1	9.0	8.0	8.8	12.7	14.5	13.9
(40) Clear-Sky Solar Irradiance	taub		0.303	0.322	0.352	0.407	0.436	0.441	0.452	0.448	0.418	0.364	0.321	0.303
	taud		2.434	2.373	2.370	2.237	2.211	2.243	2.212	2.219	2.268	2.419	2.495	2.472
	Ebn at Noon		268	280	284	274	267	266	261	258	258	261	260	259
	Edh at Noon		24	30	35	43	45	43	44	43	38	29	23	22
(44) All-Sky Solar Radiation	RadAvg		506	780	1135	1466	1672	1918	1932	1683	1393	917	582	423
	RadStd		56	102	86	120	140	127	95	98	118	99	59	47

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (60 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page